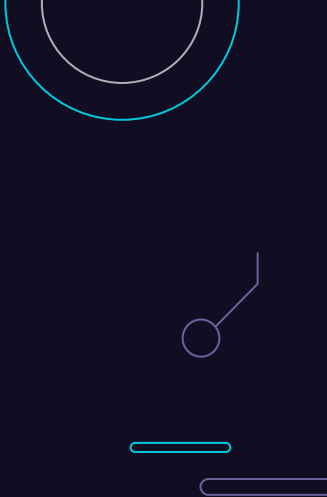




Cache Me Outside

Guanlin Li, Carolyn Liu, Lawrence Han,
Venkatesh Iyer, & Yousef Maitah

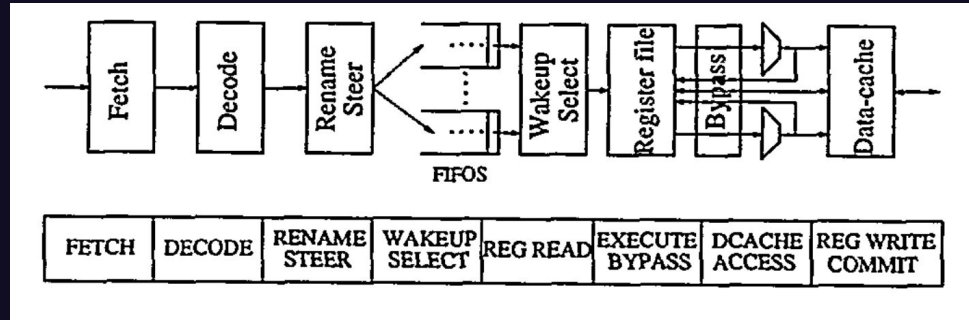


High Level Design Decisions

- R10K processor
- Palacharla RS
- N-Way Superscalar**
- Early Tag Broadcast
- Fetch enhancements
 - Tournament Branch Predictor
 - PAg
 - GSHARE
 - Return Address Stack
- Memory hierarchy improvement
 - Instruction prefetching
 - Multi Banked Caches
 - Both I & D Caches
 - Non-Blocking Caches
 - Both I & D Caches
 - Set Associative Caches
 - Both I & D Caches
 - Write Back Buffer

► Palacharla Reservation Station

- Paper by Subbarao Palacharla from the University of Wisconsin Madison
- During dispatch, all queues are searched through to check for dependencies, if none, random, else stall
 - Selects a queue using psel to insert an instruction at queue tail
 - Psels match ready queue heads with free FU
 - When a unit is assigned work, the head is popped
- 15-20% slower dispatch logic
- Issue logic is 20-30% faster
- Ideal if current critical path is RS dispatching instructions
- Optimal number of queues == number of FUs



ROB							
ht	#	Insn	R	V	S	X	C
	1						
	2						
	3						
	4						
	5						
	6						
	7						
	8						
	9						
	10						

Map Table	
Reg	T+
r1	
r2	
r3	
r4	

CDB	
T	V

RS Left FIFO					
tail	T	T1	T2	V1	V2

RS Right FIFO					
tail	T	T1	T2	V1	V2

Head

@Issue

@Issue

► N-Way

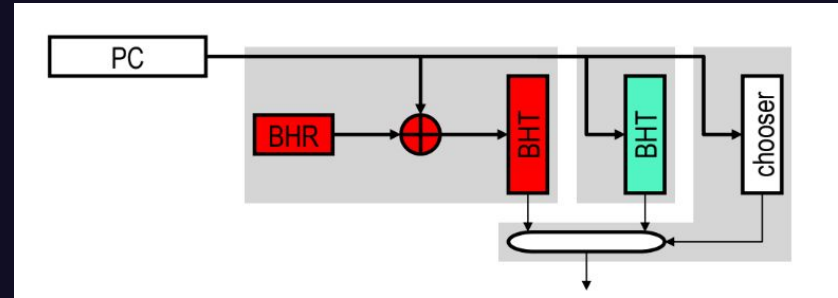
“Ain’t no way but N-Way” - A wise man

- N-way was integrated into all our components at the start of our project
- Used genvar loops to execute the same process a variable amount of times
- Improved performance of CPU a moderate amount
- Caused some issues as most testing was done on $N = 4$

► Direction Prediction

Tournament (Hybrid) Branch Predictor

- Modeled after McFarling paper
- Selects between local 2-bit correlated predictor and Gshare
- Selector is 2-bit saturating counter between local and Gshare predictors



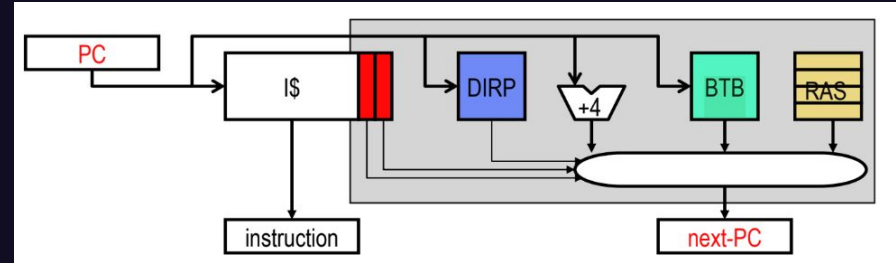
► Target Prediction

Branch Target Buffer (BTB)

- Directed mapped cache for target address of branch/jump instns
- Tag determined by upper 8 bits of PC

Return Address Stack (RAS)

- Stack that predicts the target address of return instructions
- Pushes the return address on a call and pops on a return



► Load Store Queue

Load Unit

- **3 stages**
 - Address calculation
 - Data forwarding
 - Data alignment
- Store tail pointer at time of dispatch
- Head passed signal
 - Check in RS and LU
- Issue when store addresses ready
- Bytes needed signal

Store Queue

- Stores placed in both SQ and RS
- SQ obtains address from CDB
- FIFO queue
- Retires in order
- Address ready bit
- Bytes stored signal

LSQ Forwarding

- Bitmask store instructions between head and load tail
- & bitmask and matching address vector
- Iterate through SQ size, use last valid matching entry
- Forward when bytes stored matches bytes needed

► Caches

Instruction-Cache

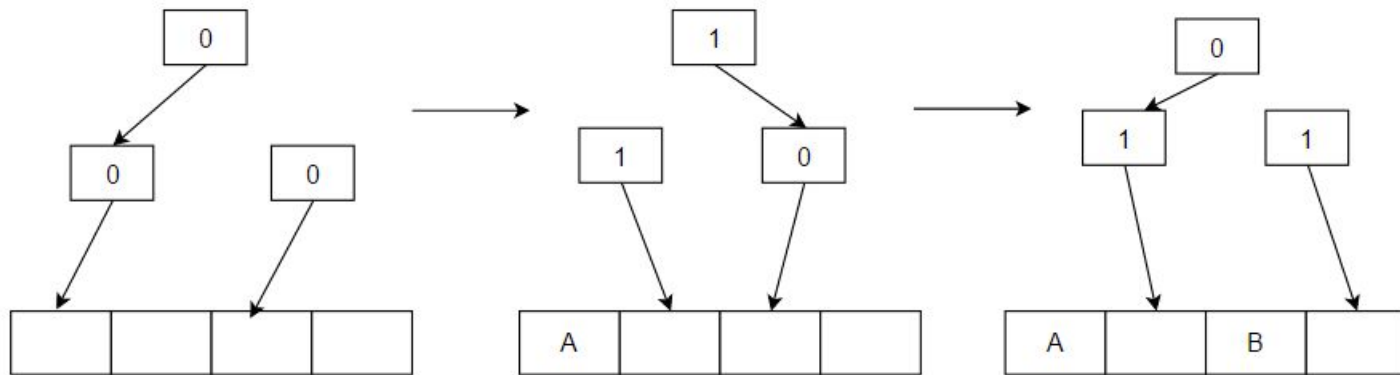
- Instruction Prefetching
- N banks

Both Caches

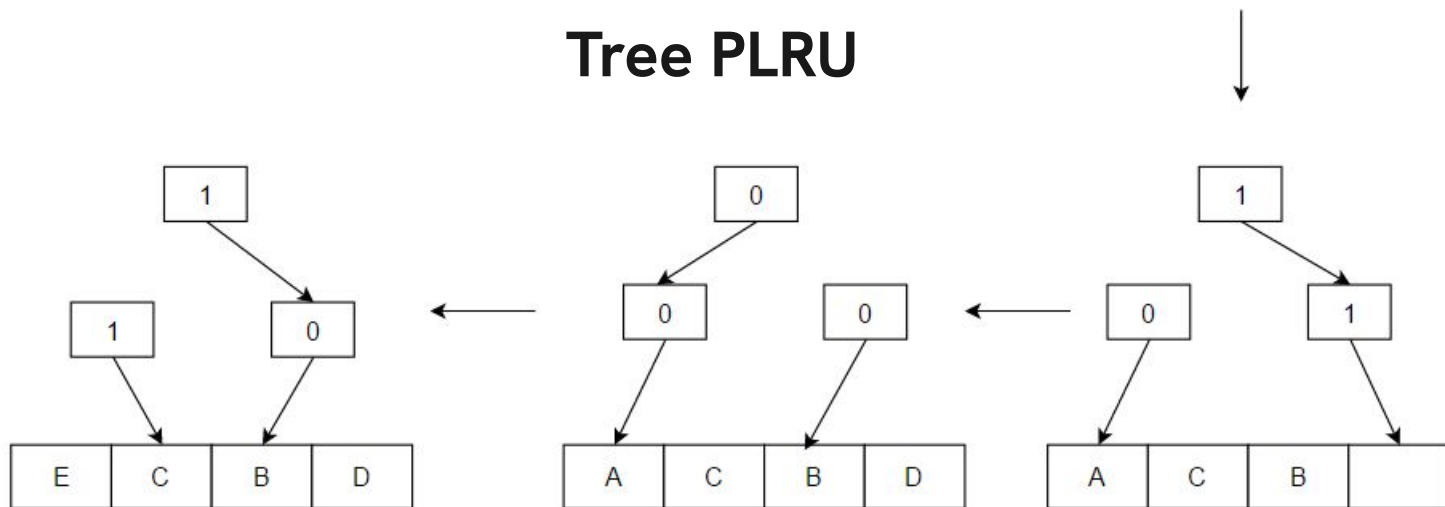
- Non-Blocking
- Configurable Multi Banking

Data-Cache

- Write Back Buffer



Tree PLRU



Performance Analysis

Optimal Parameters

N = 4;

NUM_FU_ALU = 2;

NUM_FU_MULT = 1;

NUM_FU_LOAD = 2;

NUM_FU_STORE = 2;

MULT_STAGES = 2;

ROB_SZ = 16;

LSQ_SZ = 4;

Critical Path

Unknown

(Currently synthesizing
with new changes)

Clock Period

23.0 ns

CPI

Alexnet: 1.41

Outer_Product: 1.56

Fibrec: 1.87

...

► Interesting Findings

- Weird critical path from head of ROB to load unit register
- Palacharla had a lower impact on our performance than we initially thought
- Adding prefetching cut CPI by half

► Clock Period & Critical Path Analysis

Undergoing testing and optimization

- Earliest versions hit a clock period of 13.2 ns
- Critical path was from head of ROB to load unit data
 - We are currently synthesizing

► Key Takeaways

- Starting earlier and meeting more frequently would've alleviated the time constraints.
- Work in subteams of 2-3 people.
- Build more testbenches, even if they're hard coded test cases (no need for golden models).
- Agree on the inputs and outputs of each module to avoid one side having to re-code their module.
- If a feature can be implemented in more than one module, make sure it's communicated properly which module that feature will be in.

Q&A